ReCIL: Rehearsal-Based Class Incremental Learning for Cross-Subject Motor Imagery Classification

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***Abstract*—*Objective*: Electroencephalography (EEG) based motor imagery (MI) is a classic brain-computer interface (BCI) paradigm, which can be used in neuro-rehabilitation and device control. Existing MI classification approaches only consider the scenario that the number of MI classes is pre-determined and fixed; when new MI classes arrive, data for all classes must be recollected and the model retrained, which is inefficient, and sometimes impossible, as training data for previous classes may be lost or inaccessible due to privacy concerns. *Methods*: This paper proposes rehearsal-based class incremental learning (ReCIL) for cross-subject MI classification, where the model learns new MI classes sequentially, and the test subjects are different from the training subjects. ReCIL uses Euclidean Alignment to reduce the inter-subject EEG data distribution shift, global-local replay to preserve the knowledge of old tasks, and dimensionality reduction to facilitate reliable similarity computation among EEG samples. *Results*: Experiments on two public MI datasets showed that ReCIL achieved good balance between plasticity and stability. *Significance*: To our knowledge, this is the first study on cross-subject class incremental learning for MI classification, offering a practical solution for incremental class expansion without retraining from scratch.**

***Index Terms*—Brain Computer Interface, Motor Imagery, Continual Learning, Active Learning**

I. Introduction

Brain-computer interfaces (BCIs) enable direct communication between the brain and external devices [1]. Motor imagery (MI) [2] is a classic control paradigm in BCIs, which imagines the movement of specific body parts (e.g., left hand, right hand, both feet, etc.) to elicit event-related synchronization/desynchronization [3], [4], without physically executing these movements.

Electroencephalography (EEG) is widely used in noninvasive BCIs because it is easy to acquire and has high temporal resolution [5]. In MI tasks, EEG signals can capture the activation levels of different motor cortex regions, which can be decoded by classification algorithms and mapped into control commands [6]. MI has been successfully used in robotic control [7], hand orthosis [8], and stroke rehabilitation [9].

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MI classification algorithms can be divided into traditional machine learning methods and deep learning methods. In traditional machine learning methods, common spatial pattern (CSP) [10], [11] is often combined with a traditional classifier, e.g., linear discriminant analysis [12] or support vector machine [13], for classification. In contrast, deep learning models, such as EEGNet [14] and CSP-Net [15], extract features from raw EEG signals and perform subsequent classification in an end-to-end manner and show reliable results. However, existing methods only consider static scenarios, where the number of MI classes is pre-determined and fixed. When new MI classes arrive, data for all classes must be recollected and the model retrained, which is inefficient, and sometimes impossible, as training data for previous classes may be lost or inaccessible due to privacy concerns.

Class-incremental learning (CIL) [16], [17] aims to continuously learn from new classes while preserving knowledge of previous classes, i.e., to develop a single model that can achieve high performance across all learned classes [18]. It can dynamically expand MI classes without recollecting data.

Rehearsal-based approaches are very effective in CIL [19], which retain some of the old task data and use them in new task training. The key is to save samples that can maximize knowledge from the old tasks. However, most rehearsal-based approaches only consider the representativeness (but not the diversity) of the retained samples, often leading to redundant samples. Active learning (AL) [20] aims to achieve maximum performance gains with a minimum number of labeled samples, which is conceptually similar to CIL.

This paper considers cross-subject CIL (CS-CIL) for MI classification, where the training data are from multiple existing subjects and the test data are from new (unknown) subjects. We focus on three main challenges in CS-CIL:

1) Distribution shift, which is mainly caused by significant individual differences in BCIs, and this degrades the performance of CS-CIL.
2) Catastrophic forgetting [21], which is intrinsic to CIL, as new classes modify the parameters of the model trained on old classes, degrading the performance on them.
3) Curse of dimensionality [22]. EEG data inherently have high-dimensionality, which undermines the ability to accurately assess the similarity among samples, and hence replay sample selection.

To address the distribution shift, we first apply Euclidean alignment (EA) [23], [24] to align the EEG data from different subjects. To minimize catastrophic forgetting, we propose global-local replay, which replays the EA matrices from previous tasks to preserve global knowledge, and select the most representative and diverse samples to retain local information for each class. To mitigate the curse of dimensionality, we use Kullback-Leibler divergence for feature dimensionality reduction. To our knowledge, this is the first study on CS-CIL for MI classification.

The remainder of this paper is organized as follows: Section II describes related works. Section III introduces our proposed algorithm. Section IV presents the experimental settings and results. Finally, Section V draws conclusions.

II. Related Works

A. MI Classification

Both conventional machine learning and deep learning have been utilized in MI classification [25].

In conventional machine learning, CSP [26] and its derivatives, e.g., sparse CSP [27], divergence CSP [28], probabilistic CSP [29], and filter bank CSP [30], are widely adopted in spatial filtering and feature extraction. During classification, linear discriminant analysis [12] and support vector machine [13] are commonly employed.

With the rapid advancement of deep learning, various neural network architectures have been developed for MI classification. Popularly data driven deep learning models include EEGNet [14], Deep ConvNet [31], Shallow ConvNet [31], FBCNet [32] and IFNet [33]. Recently, we proposed CSP-Net [15], a knowledge-data fusion approach to embed the CSP filtering knowledge into the data-driven neural network for improved performance. Additionally, MIRepNet [34] has been introduced as a foundation model for the MI paradigm. It proposes a high-quality data construction pipeline and an efficient pretraining strategy, demonstrating strong generalization capabilities to new devices and unseen subjects.

All existing methods for MI classification focus on static scenarios, where the MI classes are pre-determined and non-extendable. Instead, this paper considers CIL for MI classification, where new classes arrive sequentially, and only a small number of training data from previous classes can be saved for model updating.

B. Catastrophic Forgetting

Catastrophic forgetting is a main challenge to continuous learning. There are three main strategies to deal with catastrophic forgetting in CIL [19]: regularization, parameter isolation, and rehearsal.

Regularization-based methods mitigate forgetting by limiting updates to important parameters. For example, learning without forgetting (LwF) [35] uses knowledge distillation to preserve old task outputs, whereas elastic weight consolidation (EWC) [36] leverages the Fisher information matrix to estimate parameter importance. Parameter isolation methods allocate task-specific parameters. For instance, hard-attention-to-the-task (HAT) [37] reduces interference between tasks by allocating distinct parameters to them, and dynamically expandable representation (DER) [38] dynamically expands the feature extractor to accommodate new tasks. However, in CIL, regularization-based methods struggle due to the difficulty of tuning optimal parameters, and most parameter isolation methods are not applicable without task identifier.

Rehearsal-based methods replay a small subset of samples from previous tasks and integrate them into new task training. Experience replay (ER) [39] stores past data randomly, whereas incremental classifier and representation learning (iCaRL) [40] adopts herding to retain representative samples from previous classes. Building on iCaRL, bias correction (BiC) [41] further addresses the class imbalance between new and old samples. However, existing sample selection methods primarily rely on random sampling or consider only the representativeness of the samples, which are sub-optimal.

Although continual learning has been widely studied, little attention has been paid to its applications in BCI. Ma *et al.* [42] proposed a graphic neural network based few-shot CIL model for EEG-based emotion recognition. Deng *et al.* [43] proposed centroid-guided ER method (CGER) for EEG-based seizure prediction. An *et al.* [44] employed adaptive pruning to allocate task-specific parameters dynamically while maintaining privacy in EEG-based seizure detection. However, no existing work has addressed CIL in MI classification.

C. Active Learning

AL aims to maximize performance gains while minimizing the number of labeled samples required [20].

Currently, mainstream AL approaches typically combine uncertainty sampling and diversity sampling [45]. Zhdanov *et al.* [46] used k-means clustering to increase the batch diversity after uncertainty based pre-filtering. Batch active learning by diverse gradient embeddings (BADGE) [47] quantifies uncertainty through gradient norm magnitudes while ensuring sample diversity by computing the gradient vector differences. Clustering uncertainty-weighted embeddings (CLUE) [45] introduces a feature uncertainty weighted clustering approach to achieve better balance between uncertainty and diversity.

Unlike traditional AL that selects *unlabeled* samples to label, in rehearsal based CIL we select *labeled* samples from previous classes to save. Considering both representativeness and diversity of these samples better represents the original EEG data distribution and hence improves generalization.

III. Methodology

This section introduces our proposed approach for cross-subject class incremental MI classification, as illustrated in Fig. 1.

A. Problem Setting

We consider CS-CIL for MI classification, where tasks arrive sequentially and each task introduces one or more new classes. The same group of S subjects participate in all T tasks. For task t ($t = 1, ..., T$) of subject s ($s = 1, ..., S$), there are $n_{t,s}$ labeled samples $D_{t,s} = \{(X^i_{t,s}, y^i_{t,s})\}^{n_{t,s}}_{i=1}$, where $X^i_{t,s}$

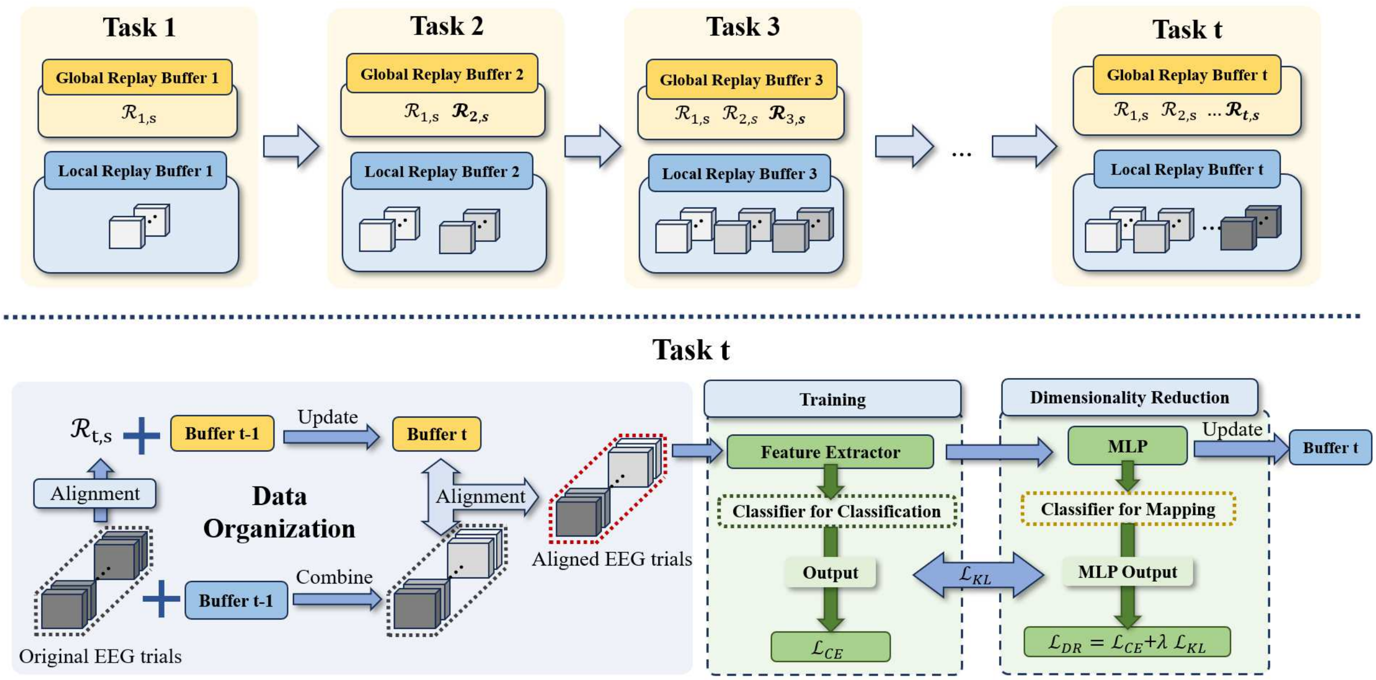

Fig. 1. Overview of the proposed ReCIL for cross-subject class incremental MI classification. For task t, new subject-specific EA matrices are computed and used to update the global replay buffer; the updated matrices then align the combined data from the current task and local replay buffer. The feature extractor is trained on EA-aligned data in two phases. After training, representative and diverse samples are selected per class via active learning to update the local replay buffer.

is the i-th EEG trial of subject s in task t, and $y^i_{t,s}$ is the corresponding class label.

A single model learns the tasks (classes) sequentially. When a new task arrives, only a very small amount of training data from previous tasks can be saved and used in model updating, due to memory constraints or privacy considerations. After training on all tasks, the model is evaluated on test data from a new subject and must classify all classes learned so far. The goal is to maintain high accuracy across all classes.

B. Overview of ReCIL

Previous work demonstrated that replaying a small number of samples from old classes can significantly mitigate catastrophic forgetting. Feature dimensionality reduction can help reduce the impact of the curse of dimensionality. Meanwhile, we [23], [24] verified that distribution shifts between different subjects can be accommodated through EA.

Inspired by these findings, our proposed ReCIL includes three key steps:

1) Data alignment, which uses EA to align data across different subjects, reducing distribution shifts in cross-subject scenarios, and enhancing knowledge sharing.
2) Global-local replay, which integrates the global replay of EA transformation matrices and the local replay of samples from previous classes.
3) Dimensionality reduction, which captures the intrinsic data relationships and facilitates more effective sample selection.

More details are explained in the following subsections.

C. Data alignment Using EA

In ReCIL, the substantial distribution shifts between different subjects lead to reduced classification accuracy. EA is adopted to address the distribution shifts and facilitate knowledge sharing between different subjects. As shown in Fig. 2, after EA, the average covariance matrices of each subject becomes an identity matrix, improving the marginal data distribution consistency across subjects.

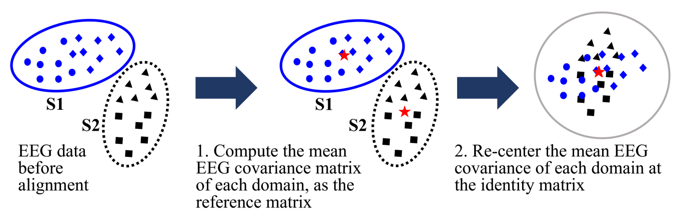

Fig. 2. EA for aligning EEG trials from different subjects.

For $n_{t,s}$ labeled samples $D_{t,s} = \{(X^i_{t,s}, y^i_{t,s})\}_{i=1}^{n_{t,s}}$ from task t ($t = 1, ..., T$) of subject s ($s = 1, ..., S$), EA first calculates the mean covariance matrix $R_{t,s}$:

$$R_{t,s} = \frac{1}{n_{t,s}} \sum_{i=1}^{n_{t,s}} X^i_{t,s} X^{i\top}_{t,s}, \tag{1}$$

and then performs the following projection for each trial:

$$\tilde{X}^i_{t,s} = R_{t,s}^{-\frac{1}{2}} X^i_{t,s}, \qquad i = 1, 2, ..., n_{t,s}. \tag{2}$$

$\{\tilde{X}^i_{t,s}\}_{i=1}^{n_{t,s}}$ then replace the original $\{X^i_{t,s}\}_{i=1}^{n_{t,s}}$ in all subsequent operations. In the following, we denote the EA reference matrix $\mathcal{R}_{t,s} \equiv R_{t,s}^{-\frac{1}{2}}$ for notation simplicity.

After EA, the average covariance matrix for each subject becomes the identity matrix:

$$\frac{1}{n_{t,s}}\sum_{i=1}^{n_{t,s}}\tilde{X}_{t,s}^{i}\tilde{X}_{t,s}^{i\top}=\frac{1}{n_{t,s}}\sum_{i=1}^{n_{t,s}}R_{t,s}^{-\frac{1}{2}}X_{t,s}^{i}X_{t,s}^{i\top}R_{t,s}^{-\frac{1}{2}}$$
$$=R_{t,s}^{-\frac{1}{2}}R_{t,s}R_{t,s}^{-\frac{1}{2}}=I. \quad (3)$$

Thus, the EEG data distribution discrepancy among different subjects is reduced.

D. Global-Local Replay

GLR contains two parts: global replay of the EA reference matrices from previous subjects and tasks, and local replay of selected samples from previous tasks.

1) Global Replay: We store the EA matrix $\mathcal{R}_{t,s}$ for each subject s and task t, computed from the current task's data only. As a new task t' arrives, to retain global knowledge and mitigate catastrophic forgetting, the global reference matrix $\bar{\mathcal{R}}_{t',s}$ is incrementally updated:

$$\bar{\mathcal{R}}_{t',s}=\frac{1}{t'}\sum_{i=1}^{t'}\mathcal{R}_{i,s}. \quad (4)$$

During training, $\bar{\mathcal{R}}_{t,s}$ aligns the combined input of current and replayed original EEG trials. At test time, the EA matrix is computed solely from the new subject's own data.

2) Local Replay: AL is commonly used in domain adaptation to select the most beneficial unlabeled samples for labeling. Its idea can also be used in CIL to better select the labeled samples to form the rehearsal set, as shown in Fig. 3 and Algorithm 1.

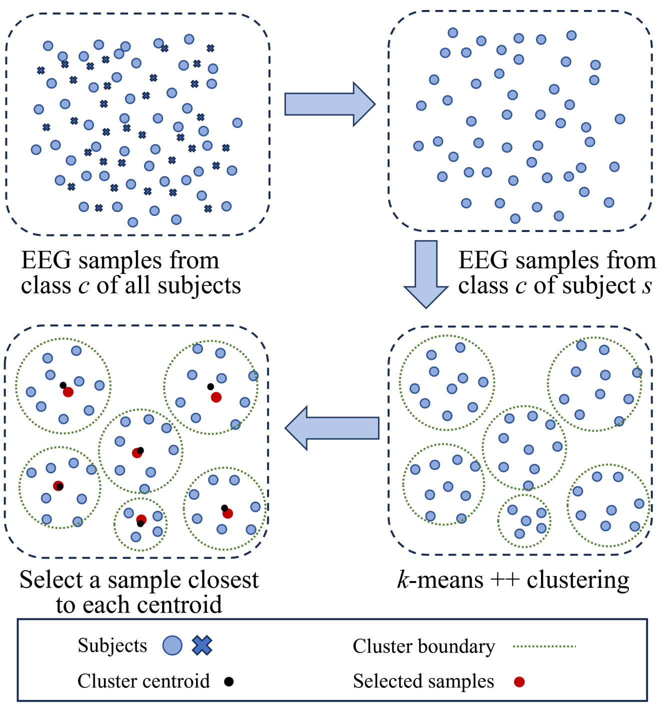

Fig. 3. Sample selection process in local replay. Perform k-means++ clustering (k equals the replay buffer size) on samples from class c of subject s, and save the sample closest to the centroid of each cluster.

Algorithm 1 The proposed local replay algorithm for class c of subject s.

Input: $\{\mathbf{x}_{c,s}^{i}\}_{i=1}^{n_{c,s}}$, $n_{c,s}$ low-dimensional EEG feature vectors from class c of subject s;
Input: M, number of replayed exemplars per class;
Input: S, number of subjects.
Output: $\mathbf{P}$, exemplar set containing the selected exemplars, initialized as an empty set for the first class.
Perform k-means++ clustering ($k=\frac{M}{S}$) on $\{\mathbf{x}_{c,s}^{i}\}_{i=1}^{n_{c,s}}$ to obtain $\frac{M}{S}$ clusters $\{\mathbf{C}_j\}_{j=1}^{M/S}$;
for $j=1$ to $\frac{M}{S}$ **do**
  Compute centroid μ_j of cluster $\mathbf{C}_j$;
  Select exemplar $\hat{X}_j$ by (5);
end for
Add $\{\hat{X}_1,...,\hat{X}_{M/S}\}$ to $\mathbf{P}$;

Given a rehearsal budget of M samples per class, and EEG samples collected from S different subjects, from each subject we select M/S samples per class, using the following procedure.

We first identify all samples belong to that specific class of the subject, and use k-means++ clustering ($k=M/S$) to partition their low-dimensional feature vectors into M/S clusters, $\{\mathbf{C}_j\}_{j=1}^{M/S}$. From each cluster, we identify the feature vector closest to its centroid, and store the corresponding original EEG trial $\hat{X}_j$ as the most representative one for rehearsal:

$$\hat{X}_j=\arg\min_{\mathbf{x}\in\mathbf{C}_j}\|\mathbf{x}-\mu_j\|_2, \qquad j=1,\dots,M/S, \quad (5)$$

where $\mathbf{C}_j$ denotes the j-th cluster, and μ_j is its centroid. In this way, we can construct a diverse and representative rehearsal set.

E. Feature Dimensionality Reduction

During the rehearsal process in CS-CIL, sample selection often relies on distances to class centers or decision boundaries, which are susceptible to the curse of dimensionality. To mitigate this, we propose KLMap, which employs a multi-layer perceptron (MLP) network and KL divergence for nonlinear dimensionality reduction. Unlike traditional linear methods such as PCA [48] or LDA [49], KLMap reduces dimensionality while preserving the discriminative power of the features.

We adopt a two-stage training process that consists of a feature extractor and an MLP, designed respectively for classification and sample selection:

1) *Classification feature learning*, where a feature extractor f and a classifier g are trained using the standard cross-entropy loss. Denote $\mathbf{z}=f(\mathbf{x})$ as the features for classification.
2) *Feature dimensionality reduction*, where a feature mapping module h and a classifier g' are trained to generate a low-dimensional representation $\mathbf{z}'=h(\mathbf{z})$ for sample selection.

The feature mapping network f is trained with loss function

$$\mathcal{L}_{\text{DR}} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{KL}}, \tag{6}$$

where λ is a hyperparameter, $\mathcal{L}_{\text{CE}}$ is the cross-entropy loss to maintain the discriminative capacity:

$$\mathcal{L}_{\text{CE}} = -\sum y_{t,s}^{i} \log\left(\text{softmax}(g'(\mathbf{z}'))^{i}\right), \tag{7}$$

and $\mathcal{L}_{\text{KL}}$ is a KL divergence loss to enforce alignment between the selection feature and the classification feature:

$$\mathcal{L}_{\text{KL}} = D_{\text{KL}}(P(\mathbf{z}) \parallel Q(\mathbf{z}')) = \sum_{i=1}^{C} P(\mathbf{z})^{i} \log \frac{P(\mathbf{z})^{i}}{Q(\mathbf{z}')^{i}}, \tag{8}$$

in which $P(\mathbf{z})$ and $Q(\mathbf{z}')$ are the softmax-normalized distributions of g and g', respectively.

Algorithm 2 gives the pseudo-code of ReCIL.

Algorithm 2 The ReCIL algorithm.

Require: Number of tasks T; number of subjects S; training datasets $D_{t,s} = \{(X_{t,s}^{i}, y_{t,s}^{i})\}_{i=1}^{n_t}$ ($t = 1, ..., T; s = 1, ..., S$); feature extractor f and the corresponding classifier g; feature mapping module h and the corresponding classifier g'; rehearsal budget M.

Ensure: Trained models f and g.

for $s = 1$ to S **do**
  Apply EA to $D_{1,s}$ to obtain $\bar{D}_{1,s}$, the aligned EEG trials;
  Store the EA reference matrix $\mathcal{R}_{1,s}$;
end for
Construct $\bar{D}^{1} = \bigcup_{s=1}^{S} \bar{D}_{1,s}$;
Train f and g on $\bar{D}^{1}$ using cross-entropy loss;
Train h and g' using $\mathcal{L}_{\text{DR}}$ on $\bar{D}^{1}$;
Use local replay on features from h to select M samples per class as the exemplar set P;
for $t = 2$ to T **do**
  for $s = 1$ to S **do**
    Apply EA to $D_{t,s}$ to obtain the EA reference matrix $\mathcal{R}_{t,s}$, and store it;
    update the EA matrix $\bar{\mathcal{R}}_{t,s}$ by (4);
    Construct $D^{t,s} = D_{t,s} \cup P_{s}$;
    Apply EA to $D^{t,s}$ to obtain $\bar{D}^{t,s}$;
  end for
  Construct $\bar{D}^{t} = \bigcup_{s=1}^{S} \bar{D}^{t,s}$;
  Train f and g on $\bar{D}^{t}$ using cross-entropy loss;
  if $t < T$ **then**
    Train h and g' on $\bar{D}^{t}$ using $\mathcal{L}_{\text{DR}}$;
    Use local replay on features from h to select M samples per class and add them to P;
  end if
end for

IV. Experiments

This section performs experiments to demonstrate the effectiveness of the proposed ReCIL. The Python code will be made available following the publication of this paper.

A. Datasets

We used two public MI datasets, BCI Competition IV-2a [50] and Weibo [51].

BCI Competition IV-2a contains EEG data from nine healthy subjects performing four MI tasks (i.e., left hand, right hand, both feet, and tongue). The EEG signals were sampled at 250 Hz. Each subject had 72 trials per class. We used the first two classes (left hand and right hand) as the first task and the last two (both feet and tongue) as the second task.

In Weibo, 10 subjects participated in experiments that included three simple limb tasks (left hand, right hand, and both feet) and three compound limb tasks (both hands, left hand with right foot, and right hand with left foot). EEG signals were resampled at 250 Hz. The experiments contained eight task sections, each included 60 trials that cover the six MI classes (10 trials/class), and one rest-state section. The six classes were sequentially divided into three tasks.

B. Preprocessing

8-30 Hz bandpass filtering was applied to both datasets to reduce artifacts and noise.

To alleviate overfitting caused by limited data in BCI Competition IV 2a, sliding windows with length 500 and overlap 25 were used to increase the number of samples.

C. Baselines

Our proposed ReCIL was compared with the following baselines:

1) Joint Training, which combines data from all tasks in training. Joint training does not suffer from forgetting and usually represents the upper bound of the classification performance.
2) Retraining, which retrains the model using only the new task data, without any strategy to prevent forgetting. This represents the lower bound of the classification performance.
3) Four regularization-based continual learning approaches, i.e., EWC [36], memory aware synapses (MAS) [52], synaptic intelligence (SI) [53], and LWF [35].
4) Three rehearsal-based continual learning approaches, i.e., ER [39], iCaRL [40], and CGER [43]. The memory buffer was set to 100 samples per class for all three approaches.

D. Experimental Settings

All experiments were conducted using the Adam optimizer with an initial learning rate of 0.001 and a batch size of 64. To address the class imbalance between replay data and new task data in both datasets, we employed the WeightedRandomSampler to ensure balanced representation across classes. We first used the last 10% of data from each subject as the validation set to determine the optimal number of epochs, and then conducted experiments using this optimal number. The maximum number of epochs was set to 60 for BCI Competition IV-2a and 70 for Weibo, with the last 10 epochs optionally used to train the feature mapping MLP. $\lambda = 0.1$

was used in the regularization-based approaches and KLMap. KLmap's dimensionality was fixed at 25. The replay budget $M = 100$ was used for every class.

Cross-subject classification was evaluated using a leave-one-subject-out protocol, where data from one subject were held out for testing, and data from all remaining subjects were used for training.

We adopted EEGNet [14] as the backbone for feature extraction and classification. Designed for general EEG recognition tasks, EEGNet comprises two convolutional blocks and a classification block. It employs temporal convolutions for frequency filtering, depthwise convolutions for spatial filtering, and separable convolutions to fuse the features, achieving both high efficiency and high performance.

The main evaluation metric was the average classification accuracy (ACC) across all tasks. We also report the performance gap between the current model and the upper bound from joint training, denoted by Δ.

All implementations were in PyTorch and conducted on a server equipped with an NVIDIA RTX 3090 GPU and 112 GB of RAM.

E. Main Experiment Results

Tables I and II present the main experiment results, with the best metrics highlighted in bold. Observe that:

1) Joint Training always achieved the best performance, as expected.
2) Although Retraining suffered from catastrophic forgetting, it still better than random guess, indicating there is transferable knowledge across tasks.
3) The four regularization-based continual learning approaches performed poorly, even worse than Retraining, because they failed to find optimal regularization strengths that balance stability and plasticity.
4) The three rehearsal-based continual learning approaches effectively reduced catastrophic forgetting by retaining task-specific data. However, their performance remained limited due to the lack of global information retention and the curse of dimensionality.
5) Our proposed ReCIL achieved the best ACC and smallest Δ. It maintained strong performance on both new and previous tasks, effectively mitigating catastrophic forgetting and distribution shift with only a few samples stored per class.

F. Ablation Study

This subsection examines the individual effect of different components in ReCIL: local replay, global replay with incremental EA, and feature dimensionality reduction.

As shown in Tables III and IV, each component contributed to the model's capacity to adapt to new tasks while retaining knowledge of previous tasks. Local replay achieved the greatest improvement by selecting a small set of representative and diverse samples that maximally capture the old task knowledge, thereby mitigating forgetting. Global replay of the EA reference matrices to reduce inter-subject distribution shift further enhanced the classification performance. Feature dimensionality reduction enables better capture of inter-sample relationships, and hence better selection of high-quality samples, further boosting ReCIL's performance. When all three components were integrated, ReCIL achieved the best performance.

G. Comparison with Other Rehearsal-Based Continuous Learning Approaches

Table V compares the performance of the local replay component in ReCIL with other rehearsal-based approaches in CIL. Each approach was evaluated with and without feature dimensionality reduction, except for Random, which is independent of feature dimensionality. Δ_1 denotes the gap to the upper bound, and Δ_2 the gap above the lower bound of old tasks. The best result within each approach is underscored, and the best overall result across all approaches is marked in bold.

The two approaches based on class centers, Class Mean and Herding, performed poorly without dimensionality reduction due to the curse of dimensionality, which hindered their ability to measure sample-to-center distances accurately. With dimensionality reduction, their performance improved significantly but still lagged behind ReCIL, as they only considered class representativeness but not diversity. Our local replay strategy achieved the best performance under the same experimental settings and replay budgets. By selecting samples that are both representative and diverse, it reduces the forgetting of old tasks while maintaining the plasticity for new tasks.

Fig. 4 visualizes the four replay methods. Observe that:

1) Compared to our replay approach, Random did not select enough samples near the decision boundary, reducing the informativeness and overall quality of the replayed samples.
2) Class Mean and Herding focus solely on the class representativeness, often selecting samples concentrated in a small region with high redundancy, failing to learn important boundary information.
3) Our proposed local replay approach selects samples that cover both the central and boundary regions of each class while reducing redundancy, better preserving class information.

H. Comparison with Active Learning Approaches

Table VI compares the performance of the local replay component in ReCIL with some sample selection approaches in AL. The number of clusters was set to 4 in all AL approaches. The best result within each approach is underscored, and the best overall result across all approaches is marked in bold.

The five AL-based selection approaches reduced forgetting by selecting the informative samples. Their performance showed little difference with and without dimensionality reduction, indicating a stronger ability to capture intrinsic data structure. However, they were still inferior to ReCIL. Specifically, Clustering is sensitive to the choice of cluster number and to initialization. CLUE and BADGE emphasize

TABLE I
EXPERIMENT RESULTS ON BCI COMPETITION IV-2A.

	Approach	Subject-specific ACC 1	2	3	4	5	6	7	8	9	Average ACC ↑	Average Δ ↓
	Joint Training	0.667	0.299	0.719	0.493	0.323	0.441	0.295	0.629	0.597	0.496	-
	Retraining	0.319	0.250	0.392	0.274	0.292	0.309	0.260	0.410	0.410	0.324	0.172
Regularization	EWC	0.347	0.292	0.295	0.288	0.267	0.302	0.319	0.351	0.309	0.308	0.188
	MAS	0.278	0.267	0.198	0.326	0.295	0.285	0.278	0.323	0.306	0.284	0.212
	LwF	0.333	0.309	0.340	0.299	0.302	0.250	0.323	0.278	0.222	0.295	0.201
	SI	0.358	0.278	0.344	0.337	0.299	0.219	0.316	0.257	0.285	0.299	0.197
Rehearsal	ER	0.483	0.260	0.451	0.281	0.330	0.313	0.319	0.462	0.618	0.391	0.105
	iCaRL	0.469	0.264	0.500	0.340	0.278	0.354	0.330	0.403	0.521	0.384	0.111
	CGER	0.396	0.236	0.379	0.278	0.319	0.281	0.309	0.427	0.451	0.342	0.154
Ours	ReCIL	0.524	0.288	0.583	0.344	0.309	0.410	0.344	0.486	0.580	**0.430**	**0.066**

TABLE II
EXPERIMENT RESULTS ON WEIBO.

	Approach	Subject-specific ACC 1	2	3	4	5	6	7	8	9	10	Average ACC ↑	Average Δ ↓
	Joint Training	0.369	0.435	0.288	0.227	0.392	0.576	0.448	0.527	0.613	0.315	0.419	-
	Retraining	0.217	0.233	0.169	0.198	0.177	0.383	0.344	0.279	0.348	0.213	0.256	0.163
Regularization	EWC	0.163	0.222	0.173	0.167	0.188	0.072	0.255	0.224	0.228	0.167	0.186	0.233
	MAS	0.156	0.186	0.169	0.163	0.188	0.177	0.190	0.192	0.154	0.179	0.175	0.244
	LwF	0.170	0.234	0.185	0.174	0.190	0.264	0.251	0.229	0.218	0.176	0.209	0.210
	SI	0.148	0.222	0.203	0.175	0.197	0.264	0.233	0.236	0.240	0.169	0.209	0.210
Rehearsal	ER	0.308	0.260	0.248	0.204	0.238	0.412	0.404	0.317	0.444	0.206	0.304	0.115
	iCaRL	0.310	0.256	0.325	0.208	0.252	0.352	0.352	0.331	0.367	0.233	0.299	0.120
	CGER	0.296	0.269	0.256	0.221	0.248	0.364	0.321	0.342	0.471	0.2021	0.299	0.120
Ours	ReCIL	0.298	0.317	0.281	0.223	0.283	0.421	0.427	0.346	0.454	0.231	**0.328**	**0.091**

TABLE III
ABLATION STUDY RESULTS OF INDIVIDUAL COMPONENTS OF RECIL ON BCI COMPETITION IV-2A.

Local Replay	Global Replay	Dimensionality Reduction	Subject-specific ACC 1	2	3	4	5	6	7	8	9	Average ACC ↑	Average Δ ↓
			0.319	0.250	0.392	0.274	0.292	0.309	0.260	0.410	0.410	0.324	0.172
✓			0.427	0.254	0.458	0.309	0.316	0.309	0.292	0.517	0.583	0.385	0.111
✓	✓		0.517	0.285	0.559	0.306	0.299	0.372	0.326	0.535	0.573	0.419	0.077
✓	✓	✓	0.524	0.288	0.583	0.344	0.309	0.410	0.344	0.486	0.580	**0.430**	**0.066**

TABLE IV
ABLATION STUDY RESULTS OF INDIVIDUAL COMPONENTS OF RECIL ON WEIBO.

Local Replay	Global Replay	Dimensionality Reduction	Subject-specific ACC 1	2	3	4	5	6	7	8	9	10	Average ACC ↑	Average Δ ↓
			0.217	0.233	0.169	0.198	0.177	0.383	0.344	0.279	0.348	0.213	0.256	0.163
✓			0.285	0.285	0.246	0.225	0.285	0.419	0.402	0.365	0.427	0.192	0.313	0.106
✓	✓		0.308	0.269	0.221	0.235	0.281	0.429	0.431	0.383	0.483	0.192	0.323	0.096
✓	✓	✓	0.298	0.317	0.281	0.223	0.283	0.421	0.427	0.346	0.454	0.231	**0.328**	**0.091**

TABLE V
COMPARISON WITH OTHER REHEARSAL-BASED CONTINUAL LEARNING APPROACHES.

Approach	Dimensionality Reduction	Subject-specific ACC 1	2	3	4	5	6	7	8	9	Average ACC ↑	Average Δ_1 ↓	Average Δ_2 ↑
Random		0.535	0.243	0.493	0.313	0.354	0.382	0.344	0.507	0.604	0.419	0.172	0.164
Class Mean	×	0.313	0.292	0.337	0.274	0.344	0.267	0.260	0.403	0.434	0.325	0.171	-0.056
	✓	0.545	0.274	0.493	0.326	0.309	0.365	0.295	0.493	0.524	0.403	0.093	0.154
Herding [40]	×	0.389	0.254	0.490	0.288	0.292	0.319	0.285	0.514	0.521	0.372	0.123	0.008
	✓	0.563	0.257	0.444	0.337	0.323	0.337	0.313	0.500	0.594	0.407	0.088	0.168
ReCIL (ours)	✓	0.524	0.288	0.583	0.344	0.309	0.410	0.344	0.486	0.580	**0.430**	**0.066**	**0.169**

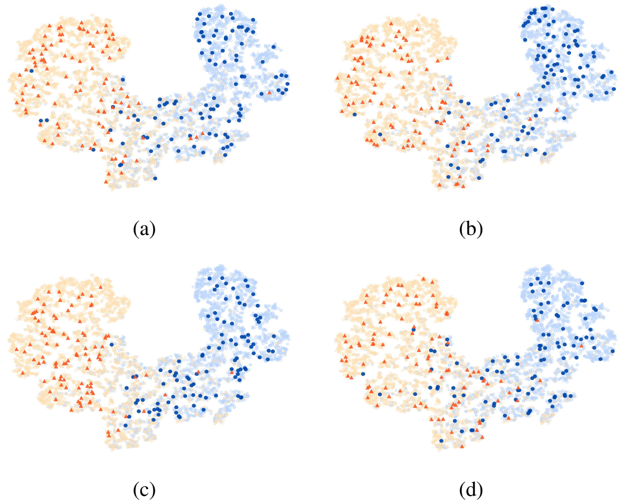

Fig. 4. t-SNE visualization of samples selected by different methods. (a) Random; (b) Class mean; (c) Herding; and, (d) our proposed local replay approach.

uncertainty for unlabeled sample selection and tended to pick outliers, which hurts the performance. GSx emphasizes diversity but not representativeness. RD is more computationally intensive yet did not surpass our approach.

I. Sensitivity Analysis

This subsection investigates the sensitivity of ReCIL to its hyperparameters, i.e., the number of replayed samples and the feature dimensionality after reduction.

Fig. 5 shows the sensitivity to the number of replayed samples. The model exhibited great plasticity when the number of replay samples was small, albeit at the expense of performance degradation on old tasks. As the number of replayed samples gradually increased, the model achieved a balance between stability and plasticity. Ultimately, all three metrics converged.

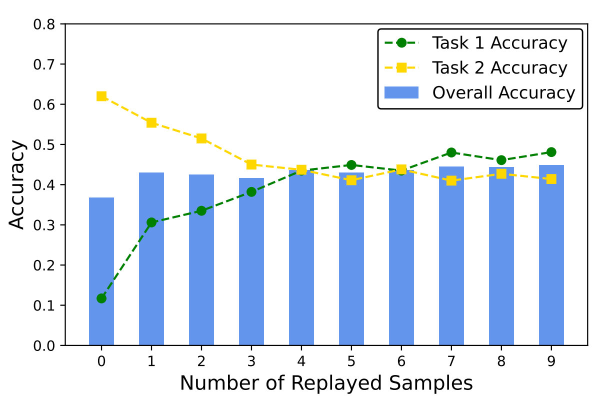

Fig. 5. Sensitivity analysis on BCI Competition IV-2a to the number of replayed samples.

Fig. 6 shows the sensitivity to the feature dimensionality after reduction. As the dimensionality increased, both the overall ACC and the old task ACC first increased and then decreased, but always better than using the original dimensionality, indicating the necessity of feature dimensionality reduction.

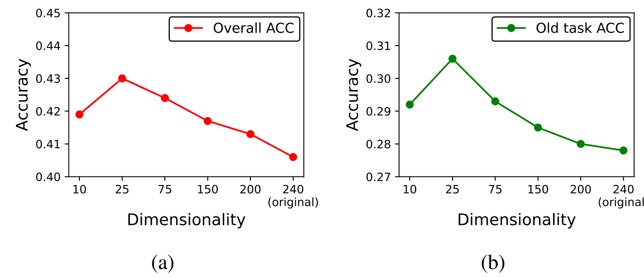

Fig. 6. Sensitivity analysis on BCI Competition IV-2a to the feature dimensionality. (a) Overall ACC; and, (b) Old task ACC.

V. Conclusions

This paper addresses the challenging scenario that new MI classes arrive sequentially, and the model needs to classify all classes, with only a very small number of training samples from previous classes saved. It has introduced ReCIL, a rehearsal based framework for cross subject class incremental learning. ReCIL uses Euclidean alignment to address the EEG data distribution shifts among different subjects, global-local replay to mitigate catastrophic forgetting, and dimensionality reduction to facilitate similarity computation for sample selection. Extensive experiments on two public MI datasets demonstrated that ReCIL achieved superior performance, offering a practical solution for incremental class expansion without retraining from scratch.

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TABLE VI
COMPARISON WITH OTHER ACTIVE LEARNING APPROACHES.

Approach	Dimensionality Reduction	Subject-specific ACC									Average ACC ↑	Average Δ_1 ↓	Average Δ_2 ↑
		1	2	3	4	5	6	7	8	9			
Clustering [54]	×	0.396	0.274	0.517	0.319	0.316	0.316	0.288	0.535	0.524	0.387	0.108	0.053
	✓	0.549	0.260	0.559	0.313	0.313	0.382	0.319	0.524	0.528	0.416	0.080	0.154
CLUE [45]	×	0.431	0.299	0.531	0.309	0.306	0.392	0.288	0.434	0.566	0.395	0.101	0.065
	✓	0.448	0.292	0.479	0.354	0.326	0.337	0.302	0.458	0.472	0.385	0.110	0.094
BADGE [47]	×	0.469	0.278	0.573	0.243	0.288	0.340	0.323	0.531	0.580	0.403	0.093	0.096
	✓	0.431	0.271	0.462	0.337	0.333	0.337	0.326	0.472	0.542	0.390	0.106	0.095
GSx [55]	×	0.504	0.264	0.497	0.351	0.323	0.319	0.302	0.458	0.497	0.390	0.105	0.104
	✓	0.368	0.254	0.500	0.267	0.326	0.337	0.267	0.476	0.486	0.365	0.131	0.054
RD [56]	×	0.479	0.288	0.559	0.340	0.299	0.323	0.333	0.566	0.542	0.414	0.081	0.143
	✓	0.504	0.309	0.517	0.337	0.340	0.372	0.333	0.420	0.538	0.408	0.088	0.164
ReCIL (ours)	✓	0.524	0.288	0.583	0.344	0.309	0.410	0.344	0.486	0.580	**0.430**	**0.066**	**0.169**

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